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Detecting viral outbreaks in future using enhanced environmental surveillance

The ongoing Coronavirus Disease 2019 (COVID-19) pandemic caused by the beta coronavirus (β CoV); Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) was first reported in Wuhan, China at the end of December 2019 (Khan et al., 2020a; World Health Organization, 2020). Genomic analysis revealed bat as a possible primary reservoir however, the intermediate host is still not clear (Nabi et al., 2020). Nonetheless, pangolins may be the possible missing link for transmission of SARS-CoV-2 from bats to humans (Zhang et al., 2020). Despite health emergency, psychological stress, and anxiety among global communities, COVID-19 has killed more than 119,400 individuals and infected more than 2 million people globally (Walker, 2020; Khan et al., 2020b).

Investigation of the COVID-19 pandemic has shown that the SARS-CoV-2 is primarily transmitted through close personal contacts especially, exposure to respiratory droplets or aerosols, sneezing, coughing, and exhalation of an infected person (Shereen et al., 2020). However, like the SARS coronaviruses, both diarrhea and virus shedding in stools have been reported for SARS-CoV-2 infected individuals (Wang et al., 2005; Xu et al., 2020; Xiao et al., 2020). In COVID-19 positive patients, a viable SARS-CoV-2 was reported in stools, where the peak viral shedding was observed one week after the onset of symptoms (Woelfel et al., 2020; Wang et al., 2020). Surprisingly, researchers in the Netherlands detected SARS-CoV-2 in the Dutch city sewage system before the first case was reported, a sign indicating the spread of COVID-19 in the local community through environmental reservoirs (Gale, 2020). Although sewage system as a route of transmission is unlikely to occur, however, the increasing number of COVID-19 positive patients in the communities will increase the concentration of SARS-CoV-2 RNA in the sewage systems (Gale, 2020). SARS-CoV-2 RNA was detected in stools from the fifth day of infection, increased progressively, and continued shedding after 30 days of infection (Yeo et al., 2020). Considering the viability of SARS-CoV-2 in the environment for days and the possibility of faecal–oral transmission may cause the spread through environmental sources; however, further extensive investigations are needed (Wu et al., 2020; Yeo et al., 2020).

Since 1989, Israel has sustained a robust sewage-based environmental surveillance system for detecting polio virus. As a result, in 2013–2014, Israel reported a silent polio epidemic in the sewage system. Immediately, vaccination was done before any cases of acute flaccid paralysis were reported (Brouwer et al., 2018). The infectious diseases like SARS-CoV-2, SARS-CoV, and MERS-CoV that are excreted *via* stools and urine in the sewage system (Wang et al., 2005; Wu et al., 2020; Yeo et al., 2020) could be used to identify the early-warning sign and scale of an outbreak. Monitoring a single treatment plant which receive influent from a huge community could provide better estimates for how widespread the COVID-19 is than testing positive individuals. This is because, the sewage surveillance system considers all those who have not been tested and have mild or no symptoms (Mallapaty, 2020). This technology could help especially, the developing countries that do not have the capabilities to screen the entire country. Similarly, in areas with no COVID-19 patients should strengthen the environmental surveillance on sewage system to identify early-warning call and react immediately. It will also be helpful to understand disease burden, increasing or decreasing pattern, and recurrent epidemic dynamics of COVID-19. Most importantly, three coronaviruses outbreaks have occurred and possibly there could be another outbreak in near future. Therefore, there is a need to establish enhanced sewage surveillance and related framework globally. This non-invasive technology could be used to limit the economic and health damage, alert the communities, and guide health officials to impose or withdraw lockdown. On the other hand, improper treatment and dumping of waste materials from the treatment centres for COVID-19 patients into the sewage water sources can lead to the spread of SARS-CoV-2 in water reservoirs. There is a chance that the virus could transmit into living flora in such water reservoirs. This could possibly increase the risk of re-emergence of the virus from sewage reservoirs and infect individuals, especially sewage workers through direct contact or vapours thus, can cause outbreaks in future. Therefore, serious attention should be given to the management and treatment of waste materials and proper treatment of environmental reservoirs with the possible presence of SARS-CoV-2.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ghulam Nabi**
Department of Cerebrovascular Diseases, The Second Affiliated Hospital of Zhengzhou University, Zhengzhou, China
Key Laboratory of Animal Physiology, Biochemistry and Molecular Biology of Hebei Province, College of Life Sciences, Hebei Normal University, Shijiazhuang, China
E-mail address: ghulamnabiqau@gmail.com.

Rabeea Siddique
Department of Cerebrovascular Diseases, The Second Affiliated Hospital of Zhengzhou University, Zhengzhou, China
Henan Medical Key Laboratory of Translational Cerebrovascular Diseases, Zhengzhou, China

Suliman Khan*
Department of Cerebrovascular Diseases, The Second Affiliated Hospital of Zhengzhou University, Zhengzhou, China
Henan Medical Key Laboratory of Translational Cerebrovascular Diseases, Zhengzhou, China
E-mail address: Suliman.khan18@mails.ucas.ac.cn.

** Corresponding author. Department of Cerebrovascular Diseases, The Second Affiliated Hospital of Zhengzhou University, Zhengzhou, China.
* Corresponding author. Department of Cerebrovascular Diseases, The Second Affiliated Hospital of Zhengzhou University, Zhengzhou, China.